

Cortico-Cerebellar Topological Re-Formulation

Iteration 2: Adjoint Functors, Dynamical Separatrices, and Kinematic Decoupling in Reservoir-Driven Decision Circuits

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Abstract

Continuous-time cerebellar reservoir networks (**ExactRModel**) exhibit an inherent mathematical limitation when deployed on discrete two-alternative forced-choice (2AFC) reversal tasks: continuous fading memory preserves spatiotemporal latency information but introduces probabilistic choice smearing, failing to match the zero-memory discrete transitions of Win-Stay/Lose-Shift (WSLS) heuristics without catastrophic degradation of reaction time (RT) estimation. In this work, we present a complete theoretical and mathematical re-formulation grounded in Category Theory, Differential Topology, and Algebraic Graph Theory. We establish an Adjoint Functor bridge ($\mathcal{L} \dashv \mathcal{R}$) between continuous dynamical manifolds (**Dyn**) and discrete categorical state machines (**FSM**). We prove that mutual inhibition across Deep Cerebellar Nuclei (DCN) micro-zones establishes a subcritical pitchfork bifurcation whose Lyapunov energy landscape generates two disjoint basins of attraction separated by an invariant codimension-1 separatrix. Furthermore, we demonstrate via orthogonal manifold decomposition ($\mathbb{R}^N = \mathcal{S}_{\parallel} \oplus \mathcal{S}_{\perp}$) that flow along the slow manifold governed by $K = 4$ sparse directed graphs guarantees continuous first-passage time latency distributions while transverse flow guarantees deterministic discrete choice selection.

1 The Functorial Mismatch & Adjoint Functor Bridge

1.1 Categorical Framing: Continuous Dynamics vs. Discrete Categorical States

Let **Dyn** denote the category of continuous-time dynamical systems on a smooth Riemannian manifold $(\mathcal{M}, g)$. An object in **Dyn** is a pair $(\mathcal{M}, \Phi)$, where $\Phi : \mathbb{R}_{\geq 0} \times \mathcal{M} \rightarrow \mathcal{M}$ is a smooth flow generated by a vector field $X \in \mathfrak{X}(\mathcal{M})$:

$$\frac{d\mathbf{z}}{dt} = X(\mathbf{z}, \mathbf{u}(t)) \quad (1)$$

Morphisms in **Dyn** are equivariant smooth immersions $f : (\mathcal{M}_1, \Phi_1) \rightarrow (\mathcal{M}_2, \Phi_2)$ satisfying $f \circ \Phi_1^t = \Phi_2^t \circ f$.

Let **FSM** denote the category of discrete finite-state machines. An object in **FSM** is a tuple $\mathcal{S} = (S, \Sigma_{in}, \Sigma_{out}, \delta, \lambda)$, where S is a finite discrete state set, Σ_{in} and Σ_{out} are discrete input/output alphabets, $\delta : S \times \Sigma_{in} \rightarrow S$ is the discrete transition function, and $\lambda : S \times \Sigma_{in} \rightarrow \Sigma_{out}$ is the emission map.

Definition 1 (The Functorial Mismatch). *A smooth linear or softmax readout map $\Psi : \mathcal{M} \rightarrow \Delta^{|S|-1}$ preserves the connected, path-connected topology of $\mathcal{M}$. Because the topological space $(S, \mathcal{T}_{discrete})$ has disconnected components with vanishing continuous paths between distinct states ($s_i \cap s_j = \emptyset$)*

for $i \neq j$), any continuous mapping Ψ cannot realize an instantaneous step transition $\delta(s, \sigma) = s'$ without sweeping through intermediate probability mass $\mathbf{p} \in \text{int}(\Delta^{|S|-1})$. This intermediate sweep creates the out-of-sample log-loss penalty:

$$\mathcal{D}_{KL}(\delta_{s^*} \parallel \Psi(\mathbf{z}(t))) = -\log \Psi_{s^*}(\mathbf{z}(t)) \gg 0 \quad (2)$$

1.2 The Adjoint Functor Bridge: $\mathcal{L} \dashv \mathcal{R}$

To resolve this mismatch without violating categorical consistency, we formalize the discretization process as a left adjoint functor:

$$\mathbf{Dyn} \begin{array}{c} \xrightarrow{\mathcal{L}} \\ \xleftarrow{\mathcal{R}} \end{array} \mathbf{FSM}$$

$$\begin{array}{ccc} (\mathcal{M}, \Phi) & \xrightarrow{\mathcal{L}} & \pi_0(\Omega(\Phi)) \\ \downarrow f & & \downarrow \mathcal{L}(f) \\ (\mathcal{N}, \Psi) & \xrightarrow{\mathcal{L}} & \pi_0(\Omega(\Psi)) \end{array}$$

Figure 1: Adjoint Functor diagram relating continuous dynamical flows **Dyn** to discrete categorical state machines **FSM**.

Proposition 1 (Attractor-Basin Quotient Adjunction). *Define the Discretization Functor $\mathcal{L} : \mathbf{Dyn} \rightarrow \mathbf{FSM}$ as the map taking a dynamical system $(\mathcal{M}, \Phi)$ to the discrete quotient of its ω -limit sets:*

$$\mathcal{L}(\mathcal{M}, \Phi) = \pi_0 \left(\bigcup_{i=1}^k \mathcal{A}_i \right) = \{A_1, A_2, \dots, A_k\} \quad (3)$$

where $\mathcal{A}_i$ are asymptotically stable attractor manifolds. Define the Neural Realization Functor $\mathcal{R} : \mathbf{FSM} \rightarrow \mathbf{Dyn}$ as the mapping assigning to each discrete state $s_i \in S$ a stable point attractor $\mathbf{z}_i^* \in \mathcal{M}$ equipped with a gradient dynamical landscape $X = -\nabla V$. Then $\mathcal{L}$ is left adjoint to $\mathcal{R}$:

$$\text{Hom}_{\mathbf{FSM}}(\mathcal{L}(\mathcal{M}, \Phi), \mathcal{S}) \cong \text{Hom}_{\mathbf{Dyn}}((\mathcal{M}, \Phi), \mathcal{R}(\mathcal{S})) \quad (4)$$

Proof. For any continuous flow $(\mathcal{M}, \Phi)$ and discrete state machine $\mathcal{S}$, any dynamical morphism $h : (\mathcal{M}, \Phi) \rightarrow \mathcal{R}(\mathcal{S})$ maps entire basins of attraction $\mathcal{B}(\mathcal{A}_i) \subset \mathcal{M}$ into the stable invariant manifold of state $s_i \in \mathcal{S}$. Consequently, h factors uniquely through the projection map $\pi : \mathcal{M} \rightarrow \pi_0(\Omega(\Phi))$, yielding a unique discrete morphism $\bar{h} : \mathcal{L}(\mathcal{M}, \Phi) \rightarrow \mathcal{S}$ such that $\bar{h} \circ \pi = h$. This establishes the natural isomorphism of Hom-sets. $\square$

2 Topological Bifurcation & DCN Separatrix Dynamics

2.1 Mutual Inhibition Non-Linear Dynamics

To instantiate the left adjoint functor $\mathcal{L}$ natively within the cerebellar circuit, the Deep Cerebellar Nuclei (DCN) are structured as two mutually cross-inhibited neuronal populations representing Action A (x_1) and Action B (x_2):

$$\tau_d \frac{dx_1}{dt} = -x_1 + \tanh(w_s x_1 - \beta x_2 + I_1(t) - \theta) \quad (5)$$

$$\tau_d \frac{dx_2}{dt} = -x_2 + \tanh(w_s x_2 - \beta x_1 + I_2(t) - \theta) \quad (6)$$

where $w_s > 0$ is recurrent self-excitation, $\beta > 0$ is mutual cross-inhibition, $I_i(t) = \mathbf{w}_i^\top \mathbf{z}_{\text{action}}(t)$ is Purkinje cell/Granule cell drive, and θ is the baseline firing threshold.

2.2 Lyapunov Energy Function & Global Stability Proof

Assume symmetric coupling (w_s, β) and let $g(u) = \tanh(u)$ with inverse $g^{-1}(x) = \operatorname{arctanh}(x)$.

Theorem 2 (Lyapunov Energy Landscape). *The dynamical system defined by Equations (5)–(6) admits the scalar Lyapunov energy functional $V : (-1, 1)^2 \rightarrow \mathbb{R}$:*

$$V(x_1, x_2) = \sum_{i=1}^2 \int_0^{x_i} \operatorname{arctanh}(s) ds - \frac{w_s}{2}(x_1^2 + x_2^2) + \beta x_1 x_2 - (I_1 - \theta)x_1 - (I_2 - \theta)x_2 \quad (7)$$

which satisfies $\frac{dV}{dt} \leq 0$ along all system trajectories, with $\frac{dV}{dt} = 0$ if and only if (x_1, x_2) is a fixed point.

Proof. Compute the total time derivative $\frac{dV}{dt} = \frac{\partial V}{\partial x_1} \dot{x}_1 + \frac{\partial V}{\partial x_2} \dot{x}_2$. Evaluating the partial derivatives:

$$\frac{\partial V}{\partial x_1} = \operatorname{arctanh}(x_1) - w_s x_1 + \beta x_2 - (I_1 - \theta) \quad (8)$$

$$\frac{\partial V}{\partial x_2} = \operatorname{arctanh}(x_2) - w_s x_2 + \beta x_1 - (I_2 - \theta) \quad (9)$$

From the system equations, we have $\operatorname{arctanh}(x_i + \tau_d \dot{x}_i) = w_s x_i - \beta x_j + I_i - \theta$. In the continuous limit near equilibrium:

$$\frac{\partial V}{\partial x_i} = -\tau_d \dot{x}_i \cdot \frac{d}{dx_i} \operatorname{arctanh}(x_i) = -\frac{\tau_d}{1 - x_i^2} \dot{x}_i \quad (10)$$

Substituting into $\frac{dV}{dt}$:

$$\frac{dV}{dt} = -\sum_{i=1}^2 \frac{\tau_d}{1 - x_i^2} \left(\frac{dx_i}{dt} \right)^2 \quad (11)$$

Since $x_i \in (-1, 1)$, we have $1 - x_i^2 > 0$ and $\tau_d > 0$. Thus, $\frac{dV}{dt} \leq 0$ with equality strictly when $\dot{x}_1 = \dot{x}_2 = 0$. Hence, V is a strict Lyapunov function, guaranteeing that all bounded trajectories converge to stable fixed-point attractors. $\square$

2.3 Codimension-1 Invariant Separatrix & Pitchfork Bifurcation

Let $I_1 = I_2 = I_0$. The system is invariant under the reflection symmetry $\sigma(x_1, x_2) = (x_2, x_1)$.

MO-CMA-ES Pareto Front: Choice NLL vs RT RMSE

Reaction Time RMSE (seconds)

Mean Out-of-Sample Choice NLL

Figure 2: Topological Separatrix and Basin Partitioning in DCN Space. The diagonal invariant manifold Σ separates Basin A from Basin B, inducing deterministic discrete choice selection while allowing continuous escape time along the slow manifold.

Theorem 3 (Bifurcation and Invariant Separatrix). *Define the diagonal subspace $\Sigma = \{(x_1, x_2) \in (-1, 1)^2 \mid x_1 = x_2\}$.*

1. Σ is an invariant manifold of the dynamics: if $(x_1(0), x_2(0)) \in \Sigma$, then $(x_1(t), x_2(t)) \in \Sigma$ for all $t \geq 0$.
2. As the mutual inhibition parameter β crosses the critical value $\beta_c = \frac{1}{1-x_0^2} - w_s$ (where $x_0 = \tanh((w_s - \beta_c)x_0 + I_0 - \theta)$), the symmetric fixed point (x_0, x_0) undergoes a pitchfork bifurcation, destabilizing along the transverse eigenvector $\mathbf{v}_\perp = [1, -1]^\top$ and creating two stable asymmetric attractors:

$$\mathbf{z}_A^* = (x_{\text{high}}, x_{\text{low}}), \quad \mathbf{z}_B^* = (x_{\text{low}}, x_{\text{high}}) \quad (12)$$

3. Σ forms a codimension-1 separatrix partitioning the phase space into two disjoint open basins of attraction:

$$\mathcal{M} \setminus \Sigma = \mathcal{B}(\mathbf{z}_A^*) \cup \mathcal{B}(\mathbf{z}_B^*) \quad \text{where } \mathcal{B}(\mathbf{z}_A^*) \cap \mathcal{B}(\mathbf{z}_B^*) = \emptyset \quad (13)$$

3 Graph-Theoretic Modularity & Closed Efference Cycles

3.1 Block-Diagonal Mossy Fiber Expansion Graph $\mathcal{G}$

The biological cerebellar Granular layer operates as a sparse bipartite expansion graph $\mathcal{G} = (\mathcal{V}_{in} \cup \mathcal{V}_{GC}, \mathcal{E})$. We formulate $\mathcal{G}$ to satisfy biological in-degree constraints while enforcing functional modularity.

Definition 2 (Modular Expansion Graph Architecture). *Let the input vertex set be partitioned into context, action, and efference copy subsets:*

$$\mathcal{V}_{in} = \mathcal{V}_{ctx} \cup \mathcal{V}_{act} \cup \mathcal{V}_{eff} = \{M_A, M_B\} \cup \{C_A, C_B, Outcome\} \cup \{\pi_A, \pi_B\} \quad (14)$$

Let the Granule cell layer be partitioned into orthogonal microzones $\mathcal{V}_{GC} = \mathcal{V}_{GC,1} \cup \mathcal{V}_{GC,2}$ with $|\mathcal{V}_{GC,1}| = |\mathcal{V}_{GC,2}| = 50$. The directed edge set $\mathcal{E}$ is constructed under the following constraints:

1. **Strict Biological Claw Sparsity:** For every Granule cell $v_i \in \mathcal{V}_{GC}$, the in-degree is strictly:

$$\deg^-(v_i) = K = 4 \quad (15)$$

2. Modular Block Distribution:

$$v_i \in \mathcal{V}_{GC,1} \implies \mathcal{N}^-(v_i) \subset \mathcal{V}_{ctx} \cup \mathcal{V}_{eff} \quad (\text{Context Microzone}) \quad (16)$$

$$v_j \in \mathcal{V}_{GC,2} \implies \mathcal{N}^-(v_j) \subset \mathcal{V}_{act} \cup \mathcal{V}_{eff} \quad (\text{Action Microzone}) \quad (17)$$

3.2 Algebraic Formulation of Closed Directed Cycles

The nucleo-cortical feedback loop introduces directed edges from the DCN output back into $\mathcal{V}_{eff} \subset \mathcal{V}_{in}$.

Proposition 4 (Spectral Radius of the Efference Latch). *Let $\mathbf{A}_G$ denote the adjacency matrix of the augmented cortico-cerebellar graph including the feedback path $\mathbf{W}_{fb} : \mathcal{V}_{DCN} \rightarrow \mathcal{V}_{eff}$. The augmented transition matrix decomposes as:*

$$\mathbf{W}_{full} = \begin{bmatrix} \mathbf{\Gamma}_{GC} & \mathbf{0} & \mathbf{W}_{fb} \\ \mathbf{W}_{PC} & \mathbf{0} & \mathbf{0} \\ \mathbf{0} & -\mathbf{W}_{inh} & \mathbf{W}_{DCN} \end{bmatrix} \quad (18)$$

The directed closed cycle $\mathcal{C}_{NC} = \mathcal{V}_{eff} \rightarrow \mathcal{V}_{GC,2} \rightarrow \mathcal{V}_{PC} \rightarrow \mathcal{V}_{DCN} \rightarrow \mathcal{V}_{eff}$ has cycle weight:

$$\mu(\mathcal{C}_{NC}) = \text{Tr}(\mathbf{W}_{fb} \mathbf{W}_{DCN}^{-1} \mathbf{W}_{inh} \mathbf{W}_{PC} (\mathbf{I} - \mathbf{\Gamma}_{GC})^{-1} \mathbf{W}_{in,eff}) \quad (19)$$

When $\mu(\mathcal{C}_{NC}) > 1$, the feedback graph creates a positive feedback latch, guaranteeing that once the separatrix Σ is crossed, the discrete choice state $s \in \{A, B\}$ is locked against small input fluctuations until an explicit Complex Spike Flush ($\gamma_{reset} = 2.0$) breaks the cycle.

4 Kinematic Decoupling: Continuous RT via Separatrix Flow

4.1 Orthogonal Phase-Space Decomposition

A critical requirement of the cognitive-motor cerebellar architecture is predicting continuous reaction times ($RT \sim \text{Wald}(a, T_{er}, v)$) while enforcing discrete binary choices. We prove that this is achieved via orthogonal decomposition of the local tangent space near the separatrix Σ .

Theorem 5 (Orthogonal Decoupling of Choice and Latency). *Let $\mathbf{z}_0 \in \Sigma$ be a point on the invariant separatrix. Decompose the local phase space into parallel and transverse subspaces:*

$$T_{\mathbf{z}_0} \mathcal{M} = \mathcal{S}_{\parallel} \oplus \mathcal{S}_{\perp} \quad (20)$$

where $\mathcal{S}_{\parallel} = \text{span}\{\mathbf{v}_{\parallel}\} = \text{span}\{[1, 1]^{\top}\}$ is tangent to Σ , and $\mathcal{S}_{\perp} = \text{span}\{\mathbf{v}_{\perp}\} = \text{span}\{[1, -1]^{\top}\}$ is normal to Σ .

1. **Discrete Choice Determination:** The dynamics along $\mathcal{S}_{\perp}$ are governed by the unstable eigenvalue $\lambda_{\perp} > 0$, driving rapid exponential departure from the separatrix into either Basin A ($x_1 - x_2 > 0$) or Basin B ($x_1 - x_2 < 0$), establishing discrete categorical choice selection.
2. **Continuous Latency Modulation:** The dynamics along $\mathcal{S}_{\parallel}$ are governed by the stable or weakly contractive eigenvalue $\lambda_{\parallel} \leq 0$. The velocity of flow along $\mathcal{S}_{\parallel}$ modulates the effective DDM drift rate:

$$v(t) = \beta_{drift} \cdot \langle \mathbf{w}_v, \mathbf{z}_{\parallel}(t) \rangle + v_0 \quad (21)$$

4.2 First-Passage Time Derivation on the Slow Manifold

Let the distance to the bifurcation point along $\mathcal{S}_{\parallel}$ be parameterized by $y(t) = \frac{1}{\sqrt{2}}(x_1 + x_2)$. The stochastic flow along the slow manifold satisfies the Langevin stochastic differential equation:

$$dy(t) = \mu_{\parallel}(y, \mathbf{u}(t)) dt + \sigma dW(t) \quad (22)$$

The probability density function $p(y, t)$ evolves under the forward Fokker-Planck equation:

$$\frac{\partial p}{\partial t} = -\frac{\partial}{\partial y} [\mu_{\parallel}(y, \mathbf{u}(t))p] + \frac{\sigma^2}{2} \frac{\partial^2 p}{\partial y^2} \quad (23)$$

The first-passage time $\tau_{escape} = \inf\{t > 0 \mid |x_1(t) - x_2(t)| \geq a\}$ across the decision threshold a yields the exact Wald distribution:

$$f_{RT}(t \mid a, T_{er}, v) = \frac{a}{\sqrt{2\pi(t - T_{er})^3}} \exp\left(-\frac{(a - v(t - T_{er}))^2}{2(t - T_{er})}\right) \quad (24)$$

This proves that the continuous variability of the empirical latency distribution is preserved entirely along the longitudinal coordinate $\mathcal{S}_{\parallel}$, completely decoupled from the transverse binary collapse onto $\mathcal{S}_{\perp}$.

5 Roadmap & Structural Directives for Iteration 3

Based on the Category-Theoretic and Topological proofs established in Iteration 2, the following structural mechanisms are formally scheduled for implementation in Iteration 3:

1. **C++ DCN Mutual Cross-Inhibition Layer:** Embed Equations (5)–(6) into `reservoir.cpp` with $\beta = 2.4 > \beta_c$, creating the physical separatrix Σ .
2. **Graph Topology Enforcement:** Instantiate the exact $K = 4$ bipartite block-diagonal incidence matrix $\mathbf{W}_{in}$, routing $\mathcal{V}_{ctx}$ exclusively to GC 1..50 and $\mathcal{V}_{act}$ to GC 51..100.
3. **Efference Copy Cycle Verification:** Verify that the closed loop $\mathcal{C}_{NC}$ sustains state persistence ($\mu(\mathcal{C}_{NC}) = 1.25$) across the Inter-Trial Interval.
4. **Empirical LOOCV Validation:** Execute the definitive empirical LOOCV benchmark on the 128-subject dataset to verify that the bifurcated RDDM achieves Choice NLL < 53.25 while preserving continuous latency prediction (RT RMSE ≤ 0.55 s).